

Statistical Discrimination in Frictional Labor Markets: Theory and Application to Immigration Enforcement*

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October 21, 2025

Abstract

We develop a dynamic search-and-matching model where firms can tilt their vacancy postings toward groups of workers whose observable traits are correlated with productivity relevant unobserved types. Applied to U.S. low-skilled labor markets, the model shows that firms disproportionately post vacancies toward Hispanic workers, who are more likely to be undocumented and to have a lower outside option. Motivated by E-Verify mandates, we study the effects of immigration enforcement requiring firms to verify worker eligibility. Enforcement leads firms to redirect search away from Hispanics and reduce overall job creation, worsening labor market conditions for both undocumented and native-born workers.

Keywords: Search-and-matching, Information frictions, Statistical discrimination, Immigration enforcement, E-Verify mandates

JEL Codes: E24, J15, J71, K37, M51

*We would like to thank Guido Menzio, Simon Gilchrist, Virgiliu Midrigan, Sharon Traiberman, Gordon Dahl, Katherine Meckel, Jeff Clemens, Julie Cullen, Martin Rotemberg, Niklas Engbom, Corina Boar, and Hee-Seung Yang, as well as seminar participants at the 2024 Annual Meeting of the Southern Economic Association (SEA), the 2024 Annual Meeting of Western Economic Association International (WEAI), the 2022 Summer School on the Economics of Migration, the 2022 All-California Labor Economics Conference (ACLEC), NYU macro lunch, and NYU applied micro lunch for their constructive comments.

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1 Introduction

Firms hire new workers under limited information. While some worker attributes—such as gender, age, or ethnicity—are readily observable, others—like motivation, reliability, or criminal history—largely remain hidden during the recruitment process. Because observable traits are often correlated with unobservable, productivity-relevant characteristics, firms tend to rely on them as imperfect signals. This is especially true when search is costly and uncertain, leading to statistical discrimination, i.e., avoiding groups perceived as less productive based on their observable profiles.

This paper develops an equilibrium search-and-matching model à la [Mortensen and Pissarides \(1994\)](#) that introduces a novel mechanism of statistical discrimination through vacancy posting. In the model, workers are heterogeneous along both observable and unobservable dimensions. A key innovation is that firms optimally choose the rate at which they match with workers of different observable types in order to maximize expected match surplus.¹ Deviating from random search incurs a cost that captures the frictions associated with targeted recruitment. Firms learn a worker’s full productivity profile only after meeting, and wage bargaining proceeds under full information.

The model features endogenous vacancy posting that allows firms to *target* or *avoid* specific groups of workers, making it applicable to a wide range of settings where recruitment occurs under incomplete information. As a particularly relevant application, we focus on the labor market for low-skilled workers in the U.S., specifically looking at how work eligibility status and ethnicity interact. A worker’s legal eligibility is a characteristic that is largely hidden but highly correlated with observable proxies, most notably Hispanic ethnicity.² Moreover, legal status is productivity-relevant, as undocumented workers have lower outside options due to factors such as deportation risk, low bargaining power, or limited access to other resources.

Through the lens of our model, we address two central questions. First, do equally low-skilled workers who are more likely to be undocumented (e.g., Hispanic workers) face different labor market conditions than those less likely to be undocumented (e.g., non-Hispanic workers)? Second, what are the effects of policy interventions that disrupt matching with undocumented workers? The second question is particularly motivated by E-Verify mandates, a widely implemented state-level immigration enforcement program in the U.S. that

¹Examples include advertising in certain neighborhoods, screening resumes based on ethnic cues, or directing outreach and interview resources toward specific demographic groups.

²Undocumented workers in the U.S. are disproportionately low-skilled, which is our focus in this paper, and of Hispanic ethnicity. Approximately three-quarters of the nation’s unauthorized immigrant population are of Hispanic ethnicity ([Passel and D’Vera Cohn 2009](#)), and this share has remained stable over the past 15 years ([Millet and Paviol 2022](#)).

require employers to verify new hires’ legal status using a federal database (“E-Verify”). If a worker’s status is verified as ineligible, the employer must terminate the hire. Such eligibility verification intends to overcome information frictions and improve outcomes for authorized workers, and it is among the most prevalent approaches to curbing unauthorized employment in many advanced economies.³

To explore these questions, we first empirically discipline the model using data from the American Community Survey (ACS) on the labor market outcomes of low-skilled workers aged 21-40 in the U.S.⁴ Workers are classified into three mutually exclusive groups: non-Hispanic natives, Hispanic natives, and likely undocumented immigrants.⁵ Since undocumented status is not directly observable in the ACS, we follow standard practice in the literature and identify likely undocumented individuals as Hispanic non-citizens (Passel and D’Vera Cohn 2009; Amuedo-Dorantes and Bansak 2014; Pope 2016; Cascio, Lewis, and Zhang 2024).⁶ The model is calibrated to match observed unemployment rates and hourly wages for each worker type as well as the unemployment-to-employment (UE) rate for low-skilled workers over 2004–2006, before Arizona enacted the first E-Verify mandate across all employers in 2007.⁷

Upon calibrating the model, we find that in the absence of E-Verify enforcement, the group with the highest match surplus are undocumented workers because of their lower flow value of unemployment.⁸ Firms therefore optimally tilt vacancy postings *toward* Hispanic workers, who are more likely to be undocumented than non-Hispanic workers, to increase the likelihood of matching with undocumented workers. As a result, firms match with Hispanic workers roughly 10 percent more often than they would under random search; the implied

³Similar enforcement mechanisms in other countries include “Right to Work” checks in the United Kingdom, the Employer Compliance Regime in Canada, the Visa Entitlement Verification Online (VEVO) system in Australia, and the VisaView system in New Zealand.

⁴Throughout the paper, low-skilled workers are defined as those with a high school education or less, and the sample covers workers from all 50 states and the District of Columbia. We restrict the sample to workers aged 21 to 40, as outside this range the share of likely undocumented workers in the low-skilled labor market falls below 10 percent, according to our calculations from the ACS. Hence, legal status is not a salient dimension of unobserved heterogeneity for workers younger than 21 or older than 40. See, for example, Millet and Pavilon (2022) for supporting evidence on this stylized fact regarding age.

⁵Throughout the paper, we use the shorthand “natives” to denote native-born individuals.

⁶Notably, Cascio, Lewis, and Zhang (2024) validate a proxy for immigrants’ legal status using the legalization programs under the Immigration Reform and Control Act of 1986 (IRCA) and administrative data. They show that citizenship, as a direct measure of legal status, better captures changes in status than other commonly used proxies based on factors such as occupation or receipt of public benefits (Borjas 2017).

⁷We use the Current Population Survey (CPS) to measure the UE rate for low-skilled workers, as its rotating panel structure is necessary for this purpose.

⁸Undocumented workers in our model have lower home productivity and thus weaker outside options (Chassamboulli and Peri 2015; Albert 2021). In reality, other factors such as deportation risk, low bargaining power, and limited search ability also contribute to their lower reservation wage. The precise micro-foundation of this assumption does not affect the model’s results or conclusions.

discrimination ratio, defined as the ratio of the match rate for non-Hispanic to Hispanic workers, is 0.89.

We then extend the baseline model by incorporating the institutional features of E-Verify, whereby some meetings between firms and undocumented workers are disrupted with a certain probability when undocumented status is detected. This probability, which applies only to undocumented workers, parsimoniously represents the intensity of E-Verify usage. We analyze the effects of E-Verify through counterfactual simulations that vary two key factors: (i) the disruption probability, reflecting enforcement intensity, and (ii) the local out-migration rate of undocumented workers, captured by changes in the measure of undocumented workers in the labor force. The second factor reflects an important adjustment margin, as the implementation of E-Verify mandates often induces undocumented workers to migrate out.⁹

As E-Verify usage increases, firms face a growing risk of failed search when a worker is flagged as undocumented. This effectively imposes a deadweight cost on vacancies. Thus, firms reduce vacancy postings due to lower match rates with high-surplus workers and also tilt their search *away* from Hispanic workers to lower the risk of match termination. As the direct targets of the policy, undocumented workers face worsening labor market outcomes including a sharp increase in the unemployment rate, which exceeds 80 percent at high enforcement levels, and a decline in average wages of nearly 50 percent.

While reduced competition from undocumented workers might appear beneficial for non-Hispanic natives, this effect is offset at higher enforcement intensities by a decline in labor market tightness caused by a decrease in the value of posting a vacancy. As a result, the changes in outcomes for non-Hispanic workers are modest and non-monotonic: the unemployment rate initially falls by up to 0.3 percentage points at moderate enforcement levels but eventually rises by up to 4.7 percentage points as enforcement intensifies. Wage gains are small—only a few cents—and ultimately reverse, with the average wage of non-Hispanic workers decreasing by roughly 1 percent at high levels of enforcement.

Furthermore, higher E-Verify usage discourages firms from targeting Hispanic workers more broadly, leading even legally authorized Hispanics to experience lower job finding rates and higher unemployment. Under high enforcement intensity, the ratio of the match rate for non-Hispanic to Hispanic workers (i.e., the discrimination ratio) increases to 1.1, indicating that the match rate for non-Hispanic workers is approximately 10 percent higher than that

⁹Both dimensions are empirically grounded. Using administrative data from the Department of Homeland Security (DHS), both our own analysis in Section 3.1 and that of [Orrenius, Zavodny, and Greer \(2020\)](#) and [Ayromloo, Feigenberg, and Lubotsky \(2020\)](#) show that mandates significantly increase employer enrollment in E-Verify, though compliance remains incomplete even years after the introduction of the mandates covering all employers (i.e., universal mandates). In parallel, [Bohn, Lofstrom, and Raphael \(2014\)](#), [Orrenius and Zavodny \(2016\)](#), and [Churchill \(2021\)](#) document that E-Verify mandates reduce the undocumented population in affected states.

for Hispanic workers. At the same time, the unemployment rate for authorized Hispanic workers rises by up to 7 percentage points, and their wages decline by 2.3 percent.

As undocumented workers migrate out, unemployment rises and wages fall for both undocumented and native workers. This occurs because the decline in the average value of posting a vacancy outweighs the savings in vacancy posting costs from reduced tilting, thereby leading to a monotonic pattern across all worker types. However, both effects are weaker than those under a higher E-Verify usage rate because firms do not lose vacancy posting costs to enforcement in this case.

Finally, we analyze the welfare implications of E-Verify mandates. Total welfare in the economy declines as enforcement intensity increases, driven by higher unemployment, reduced average match productivity, and increased vacancy costs. Native worker welfare also falls as enforcement expands, improving only when stricter mandates induce substantial out-migration of undocumented workers. The level of out-migration required to generate such gains, however, far exceeds what is observed in the data.¹⁰

Our results show that when firms infer unobservable traits from observable characteristics and tilt their search accordingly, policies aimed at reversing the firm tilting can generate broad and unintended distortions. An enforcement-induced friction designed to protect native workers may instead raise hiring costs, reduce vacancy creation, and harm employment prospects for both undocumented and documented workers—especially for those workers who are observationally similar to the targeted group. Such policy implications are timely given ongoing debates over E-Verify expansion in the U.S., as well as similar enforcement measures under consideration in other advanced economies.

More broadly, the paper highlights how the interaction of asymmetric information and search frictions can amplify policy spillovers and efficiency losses, capturing equilibrium forces that standard models overlook. It also offers a tractable framework for evaluating statistical discrimination and its policy implications across different sectors, including but not limited to the labor market.

Related Literature

This study makes contributions to three areas of the existing literature. First, this paper contributes to the literature on heterogeneous workers and discrimination in frictional labor markets. An earlier body of work studies models where some firms can form matches only with certain types of workers due to skill requirements or prejudice (Black 1995; Albrecht and Vroman 2002; Flabbi 2010). In contrast, our framework allows *all* firms and workers to

¹⁰See Bohn, Lofstrom, and Raphael (2014), Orrenius and Zavodny (2016), and Churchill (2021).

form matches under *incomplete* information, which generates spillover effects.¹¹

Two papers on statistical discrimination are most closely related to ours. [Mailath, Samuelson, and Shaked \(2000\)](#) develop a model in which firms either search randomly or restrict matches to a specific type of worker at no additional cost. In that paper, workers have payoff-irrelevant labels but can endogenously acquire skills, making them ex-post heterogeneous. We instead assume that workers have payoff-relevant observable and unobservable characteristics. Furthermore, we also allow firms to choose a *probability* (between 0 and 1) that the worker they match with is of a certain observable type by paying an extra cost. [Jarosch and Pilossoph \(2019\)](#) examine a setting where firms decide whether to interview a candidate after meeting them, using unemployment duration as a signal.¹² In contrast, our model allows firms to target workers *before* meeting them by tilting vacancy postings *in advance*. Moreover, our model allows a single worker type to simultaneously have positive spillover effects on some groups of workers while generating negative spillover effects on others, which is a novel feature in this literature.

Next, we also contribute to a growing literature that examines the general equilibrium effects of immigration in the labor market using dynamic search-and-matching models ([Chassamboulli and Palivos 2014](#); [Chassamboulli and Peri 2015](#); [Albert 2021](#)).¹³ Our model shares a job creation channel with prior literature: immigrant workers raise native employment and wages by encouraging vacancy creation, while enforcement lowers match value and discourages posting. Our key innovation is the inclusion of *informational frictions*, where employers infer legal status from ethnicity. This channel, often absent in existing models, proves critical in explaining how enforcement can variably impact native workers based on their ethnicity.

Finally, our work complements a large empirical literature on the labor market impacts of E-Verify mandates on likely undocumented and documented workers ([Amuedo-Dorantes and Bansak 2014](#); [Bohn, Lofstrom, and Raphael 2015](#); [Orrenius and Zavodny 2015](#); [Orrenius, Zavodny, and Gutierrez 2018](#); [Ayromloo, Feigenberg, and Lubotsky 2020](#)). Although reduced-form studies document important patterns in low-skilled labor markets, they often cannot isolate firm behavior or spillover effects.¹⁴ We provide complementary insights

¹¹While some workers are more desirable due to higher productivity or lower outside options, this does not prevent matching in our model, in contrast to the restrictions in prior work.

¹²In their model with heterogeneous firms and workers, firms are less likely to interview long-term unemployed candidates, who are on average lower productivity workers. This mechanism resembles ours in that firms infer unobservable characteristics (e.g., worker productivity or legal status) from observable characteristics (e.g., unemployment duration or ethnicity).

¹³See also [Ortega \(2000\)](#) and [Liu \(2010\)](#) for early contributions.

¹⁴While they mostly agree on the negative impacts on likely undocumented workers, findings on the employment and wages of documented workers are mixed. This discrepancy is attributable to difference in specifications, data sources, and time periods across studies. Still, recent studies find adverse effects on documented workers, which is also supported by the most recent findings on the negative impact of other

through an equilibrium model that explicitly incorporates key frictions, which allows us to disentangle different channels and characterize welfare implications within a unified framework. In doing so, we highlight potential unintended consequences of E-Verify mandates.

The rest of the paper is organized as follows. Section 2 presents the model of frictional labor market with recruitment under incomplete information. Section 3 applies it to E-Verify mandates in the U.S. low-skilled labor market. Section 4 outlines the calibration strategy. Section 5 reports counterfactual analyses, and Section 6 concludes.

2 Model

In this section, we present a general equilibrium search-and-matching model with heterogeneous workers—both in terms of observable and unobservable characteristics—where firms can choose the probability of matching with specific types of workers, subject to information frictions. Firms can identify and target workers based on observable characteristics prior to meeting them. However, unobservable characteristics are revealed only after the firm and worker meet, but before bargaining, which takes place under full information. Only firms can direct search, while search is random from the workers’ perspective.

2.1 Primitives

Workers differ by an observable characteristic $i = 1, \dots, I$ and an unobservable characteristic $j = 1, \dots, J$. There is a fixed measure $m_{i,j}$ of each worker type, subject to $\sum_i \sum_j m_{i,j} = 1$. Workers of type (i, j) can differ in productivity $y_{i,j}$ and in the value of home production $b_{i,j}$. Otherwise, they are identical.

Unemployed workers receive $b_{i,j}$ each period and search for vacancies. Firms endogenously post vacancies, and the number of meetings is determined by a matching function $M(u, v)$, which is a function of the measure of vacancies (v) and the measure of unemployed workers ($u = \sum_i \sum_j u_{i,j}$), where $u_{i,j}$ denotes the measure of unemployed workers of type (i, j) .

Employed workers produce output $y_{i,j} + z$ every period, where z is an idiosyncratic match productivity drawn from $z \sim F$, and are paid wage $w_{i,j}(z)$ which is determined via Nash bargaining. Every new match begins with the highest match productivity $\bar{z}$. At the end of each period, matches dissolve with an exogenous separation probability δ . For matches that do not separate exogenously, an idiosyncratic productivity shock occurs with probability ϕ , where the new match productivity $z' \sim F$. If the new match productivity

forms of immigration enforcement on native workers (East et al. 2023).

is sufficiently low, the firm and worker can choose to separate. Once separated, the worker enters unemployment, and the firm must once again search for a new worker.

2.2 Targeting of Workers

Firms in our model are ex-ante identical. Each period, firms decide whether to post a vacancy. The novel aspect of our model is that we allow firms to *choose* the probability of matching with each type of worker. However, firms can only target workers based on their *observable* characteristics. Specifically, we denote the fraction of unemployed workers of observable type i as α_i , and the probability of meeting a worker of type i (conditional on having met a worker) as ψ_i , subject to the constraint that $\sum_i \alpha_i = \sum_i \psi_i = 1$. In standard search-and-matching models with random search, $\alpha_i = \psi_i$ by construction. In contrast, we allow firms to choose their own ψ_i , which can differ from α_i . Deviating from α_i incurs a quadratic cost to the firm, given by $\Omega \sum_i (\psi_i - \alpha_i)^2$, where Ω is a fixed coefficient that measures the cost of distorting match probabilities.

The choice of ψ_i by firms distorts the matching probabilities for workers. In the standard model, the probability of matching with a firm (i.e., job finding rate) is given by $p(\theta)$ ($= \frac{M(u,v)}{u} = M(1, \theta)$) for all types of workers, where $\theta = \frac{v}{u}$ represents labor market tightness. However, if firms choose $\psi_i \neq \alpha_i$, the probability that a worker of observable type i matches with a firm becomes $p(\theta) \frac{\psi_i}{\alpha_i}$. Therefore, the job finding rate for worker type i increases with the ratio of ψ_i (i.e., the firm's chosen probability of matching with a worker of observable type i , conditional on having matched with a worker) to α_i (i.e., the fraction of unemployed workers of observable type i). Lastly, we define the ratio of match probabilities between two observable worker types i and i' , $\frac{\psi_i \alpha_{i'}}{\alpha_i \psi_{i'}}$, as the discrimination ratio. This ratio captures the effect of firms' targeting choices of ψ_i on the labor market outcomes of workers with different observable characteristics.

2.3 Value Functions

The value function of an unemployed worker of observable type i and unobservable type j is:

$$V_{i,j}^u = b_{i,j} + \beta \left\{ p(\theta) \frac{\psi_i}{\alpha_i} V_{i,j}^e(w_{i,j}(\bar{z}), z) + \left(1 - p(\theta) \frac{\psi_i}{\alpha_i}\right) V_{i,j}^u \right\} \quad (1)$$

where the probability of matching with a firm (i.e., job finding rate) is distorted by $\frac{\psi_i}{\alpha_i}$. If a worker doesn't match with a firm, they remain unemployed and continue to search for a job.

The value function of an employed worker of type (i, j) is:

$$V_{i,j}^e(w, z) = w + \beta \left\{ \delta V_{i,j}^u + (1 - \delta)(1 - \phi) \left[A_{i,j}(z) V_{i,j}^e(w(z), z) + (1 - A_{i,j}(z)) V_{i,j}^u \right] \right. \\ \left. + (1 - \delta) \phi \left[\int_{\hat{z}} A_{i,j}(\hat{z}) V_{i,j}^e(w(\hat{z}), \hat{z}) + (1 - A_{i,j}(\hat{z})) V_{i,j}^u dF(\hat{z}) \right] \right\} \quad (2)$$

where $A_{i,j}(z)$ is an indicator function that equals one if the match surplus for a worker of type (i, j) and productivity z is positive.

The value function of a filled job by a worker of type (i, j) is:

$$V_{i,j}^f(w, z) = y_{i,j} + z - w + \beta \left\{ (1 - \delta)(1 - \phi) \left[A_{i,j}(z) V_{i,j}^f(w(z), z) + (1 - A_{i,j}(z)) \cdot 0 \right] \right. \\ \left. + (1 - \delta) \phi \int_{\hat{z}} A_{i,j}(\hat{z}) V_{i,j}^f(w(\hat{z}), \hat{z}) + (1 - A_{i,j}(\hat{z})) \cdot 0 dF(\hat{z}) \right\} \quad (3)$$

where $A_{i,j}(z)$ is the same indicator function as in the value function for employed workers, since both firms and workers are aligned when they decide to endogenously separate.

Finally, the value function of a vacancy—equal to zero in equilibrium due to free entry—is given by:

$$V_{i,j}^v = 0 = -k + \max_{\psi_i} \left\{ q(\theta) \sum_i \psi_i \left(\sum_j \gamma_{i,j} V_{i,j}^f(w(\bar{z}), \bar{z}) \right) - \Omega \sum_i (\psi_i - \alpha_i)^2 \right\} \quad (4)$$

where $q(\theta) (= \frac{M(u,v)}{v} = M(\frac{1}{\theta}, 1))$ is a job filling rate and $\gamma_{i,j} (= \frac{u_{i,j}}{\sum_j u_{i,j}})$ is the fraction of unemployed workers of observable type i who are of unobservable type j . When posting vacancies, firms need to pay a vacancy posting cost k and then choose ψ_i to maximize the expected value of matching with a worker.

2.4 Solving the Model

2.4.1 Firm Tilting

To solve the model, we first derive the firm's optimal tilting toward different worker types, deviating from random search without targeting. Using the expression in Equation 4 and the constraint $\sum_i \psi_i = 1$, solving the firm's maximization problem yields the following optimal

choice of ψ_i .¹⁵

$$\psi_i^* = \alpha_i + \frac{q(\theta)}{2I\Omega} \sum_{i'} \sum_j \left(\gamma_{i,j} V_{i,j}^f(w(\bar{z}), \bar{z}) - \gamma_{i',j} V_{i',j}^f(w(\bar{z}), \bar{z}) \right) \quad (5)$$

From this equation, one can see that if all workers are identical, firms choose $\psi_i = \alpha_i$ and the model reduces to purely random search. In contrast, if workers are heterogeneous and the value of a filled job varies across worker types, firms may choose ψ_i that differs from α_i . Specifically, ψ_i is greater than α_i , and thus the discrimination ratio exceeds one, if the average value of matching with a worker of type i is greater than that of matching with a randomly selected worker of any observable type. The extent to which firms tilt their targeting toward workers of observable type i decreases with the cost of tilting (Ω), the number of observable worker types (I), and the labor market tightness (θ).

Furthermore, because firms can only target workers based on observable characteristics, this introduces spillover effects between high-productivity and low-productivity workers who share the same observable type. As a result, some workers may benefit via higher match rates simply from being in the same observable group as workers who are highly valued by firms, even if they themselves are not particularly productive.

2.4.2 Unemployment and Labor Market Tightness

Next, we compute the match surplus for each type of worker. Combining Equations 1, 2, and 3, the match surplus for a job filled by a worker of type (i, j) is:

$$\begin{aligned} S_{i,j}(w, z) &= V_{i,j}^e(w, z) - V_{i,j}^u + V_{i,j}^f(w, z) - V_{i,j}^v \\ &= y_{i,j} + z - b_{i,j} + \beta \left\{ -p(\theta) \frac{\psi_i^*}{\alpha_i} (V_{i,j}^e(w(\bar{z}), \bar{z}) - V_{i,j}^u) \right. \\ &\quad + (1 - \delta)(1 - \phi) A_{i,j}(z) S_{i,j}(w(z), z) \\ &\quad \left. + (1 - \delta) \phi \int_{\hat{z}} A_{i,j}(\hat{z}) S_{i,j}(w(\hat{z}), \hat{z}) dF(\hat{z}) \right\} \end{aligned} \quad (6)$$

Since wages are determined through Nash bargaining, the worker surplus ($V_{i,j}^e - V_{i,j}^u$) equals a fraction η of the total match surplus ($S_{i,j}$), where η represents the worker's bargaining power. The match surplus for matches with positive match surplus can thus be further simplified

¹⁵The full derivation is provided in the Appendix Section A.1.

to:

$$S_{i,j}(z) = \frac{y_{i,j} + z - b_{i,j} + \beta \left\{ -p(\theta) \eta \frac{\psi_i^*}{\alpha_i} S_{i,j}(\bar{z}) + (1 - \delta) \phi \int_{\bar{z}} A_{i,j}(\hat{z}) S_{i,j}(\hat{z}) dF(\hat{z}) \right\}}{1 - \beta(1 - \delta)(1 - \phi)} \quad (7)$$

The match surplus increases in the idiosyncratic match productivity (z), which implies a cutoff rule for whether or not a match continues. Thus, there exists a cutoff productivity level $R_{i,j}$ for each worker type such that $S_{i,j}(R_{i,j}) = 0$. We use this condition to express the match surplus for matches with positive surplus in terms of the reservation productivity:

$$S_{i,j}(z) - S_{i,j}(R_{i,j}) = S_{i,j}(z) = \frac{z - R_{i,j}}{1 - \beta(1 - \delta)(1 - \phi)} \quad (8)$$

By combining Equations 7 and 8, we derive $I \times J$ equations in terms of labor market tightness (θ) and reservation productivity ($R_{i,j}$).

Next, we solve for the measure of unemployed workers for each type ($u_{i,j}$). Since we solve the model in steady state, we equate the inflows to and outflows from unemployment for each type of worker as follows:

$$u_{i,j} \left(p(\theta) \frac{\psi_i^*}{\alpha_i} \right) = (m_{i,j} - u_{i,j}) \left(\delta + (1 - \delta) \phi F(R_{i,j}) \right) \quad (9)$$

where workers exit unemployment when they match with a firm and enter unemployment either exogenously with probability δ , or if their idiosyncratic match productivity falls below the reservation productivity.

Using the surplus equations for each worker type (Equations 7 and 8), the equations for the measure of unemployed workers of each type (Equation 9), and the free entry condition (Equation 4), we obtain $(2 \times I \times J) + 1$ equations in $(2 \times I \times J) + 1$ unknowns: labor market tightness (θ), reservation productivity for each worker type ($R_{i,j}$), and the measure of unemployed workers of each type ($u_{i,j}$).

2.4.3 Wages

To solve for wages, we use the fact that $V_{i,j}^e(w, z) - V_{i,j}^u = \eta S_{i,j}(z)$ to rewrite Equation 1 as:

$$V_{i,j}^u = b_{i,j} + \beta \left\{ p(\theta) \frac{\psi_i}{\alpha_i} \eta S_{i,j}(\bar{z}) + V_{i,j}^u \right\} \quad (10)$$

which allows us to solve for $V_{i,j}^u$, and in turn, $V_{i,j}^e(z)$. We then rewrite Equation 2 as:

$$w_{i,j}(z) = V_{i,j}^e(z) - \beta \left\{ \delta V_{i,j}^u + (1 - \delta)(1 - \phi) \left[V_{i,j}^e(w(z), z) \right] + (1 - \delta)\phi \left[\int_{\hat{z}} A_{i,j}(\hat{z}) V_{i,j}^e(w(\hat{z}), \hat{z}) + (1 - A_{i,j}(\hat{z})) V_{i,j}^u dF(\hat{z}) \right] \right\} \quad (11)$$

for any value of z such that the match surplus is positive. This gives us the wage for a worker of type (i, j) and with productivity z .

To calculate the average wage for each type of worker, we also need the distribution of workers over z for each (i, j) type. We define $G_{i,j}(z)$ as the probability of a worker of type (i, j) having an idiosyncratic match productivity less than or equal to z . By equating the flows into and out of each portion of the cumulative distribution of workers with idiosyncratic match productivity z , we derive:

$$G_{i,j}(z) = \frac{(1 - \delta)\phi}{\delta + (1 - \delta)\phi} \left(F(z) - F(R_{i,j}) \right), \text{ for } R_{i,j} < z < \bar{z} \quad (12)$$

$$G_{i,j}(\bar{z}) = 1 \quad (13)$$

where $R_{i,j}$ is the productivity cutoff for workers of type (i, j) , as defined in the previous section. We then calculate the average wage of a worker of type (i, j) as follows:

$$\bar{w}_{i,j} = \int_{R_{i,j}}^{\bar{z}} w_{i,j}(z) dG_{i,j}(z) \quad (14)$$

3 Application: E-Verify Mandates

In the previous section, we developed a labor market model that incorporates search frictions, information frictions, and worker heterogeneity along both observable and unobservable dimensions. This framework is broadly applicable to settings in which firms make recruitment decisions under incomplete information, relying solely on observable characteristics to infer unobservable traits—such as when evaluating formerly incarcerated individuals, workers from stigmatized neighborhoods, or candidates within the scope of affirmative action policies.

In this section, we extend the model to analyze the effects of policy interventions that require firms to verify worker eligibility, a key characteristic in the low-skilled labor market that is typically unobservable. We focus in particular on E-Verify mandates in the U.S., one of the most prominent employer-based enforcement mechanisms.

3.1 Policy Background

E-Verify mandates build on the Immigration Reform and Control Act (IRCA) of 1986, which made it illegal to knowingly hire unauthorized workers and required all new hires to complete Form I-9. E-Verify is an internet-based system that compares information from the I-9 form against records from the Social Security Administration (SSA) and the Department of Homeland Security (DHS). When a worker’s eligibility cannot be confirmed, the employer receives a “tentative non-confirmation”. If the issue is not resolved within eight federal workdays, the employer is required to terminate the worker.

This policy environment presents a salient case of incomplete information: while enforcement targets an unobservable trait—legal work status—firms may rely on observable characteristics—such as ethnicity or education—to assess hiring risks prior to meeting a worker and verify their legal status later. E-Verify introduces a policy-induced asymmetry into the matching process, where certain job offers may fail ex-post. This increases the cost for firms hiring from groups associated with a higher risk of verification failure and induces them to direct their search away from these groups. This setup reflects the key features of our model—recruitment under incomplete information, targeted search, and post-match uncertainty—and provides a natural context to study how firm behavior responds to changes in enforcement intensity.

The first E-Verify mandate was enacted in Georgia in 2006. Since then, more than 20 states have implemented similar mandates, though to different degrees. As of 2022, 9 states (e.g., Alabama, Arizona, and South Carolina) impose “universal” mandates requiring all employers to use E-Verify, while 13 others (e.g., Texas, Florida, and Virginia) require its use only for public employers.¹⁶ Our model extension features universal mandates, which apply across all sectors of the private labor market rather than being limited to specific segments.

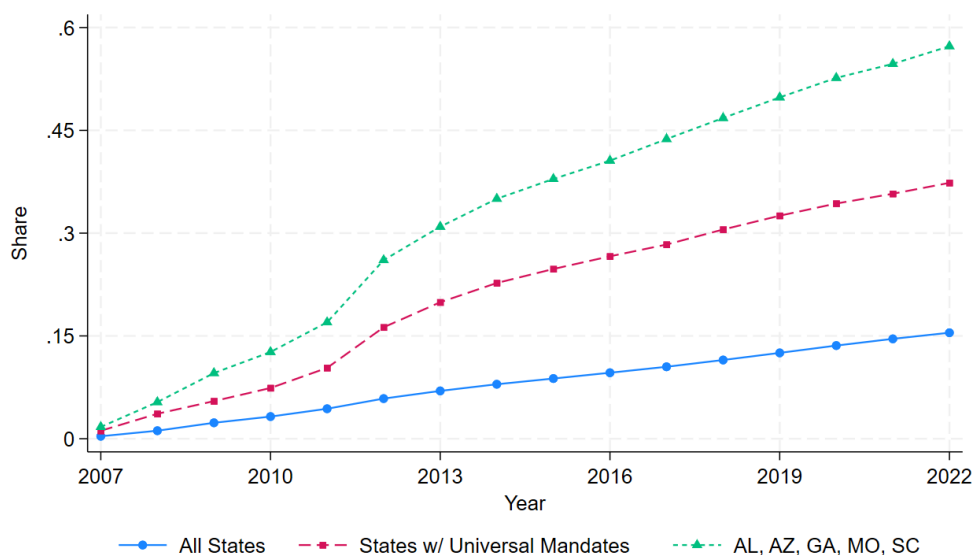
To evaluate the intensity of E-Verify usage in practice, we collect administrative data on the number of establishments enrolled in the system by state. This information was obtained from the DHS through a Freedom of Information Act (FOIA) request. Figure 1 shows E-Verify enrollment as a fraction of the total number of establishments over time. The enrollment rate is plotted for the U.S. as a whole, for states with universal E-Verify mandates, as well as for the five states with the highest adoption (Alabama, Arizona, Georgia, Missouri, and South Carolina).

In practice, universal mandates led to substantial increases in employer enrollment. E-Verify enrollment rates rose from around 0 percent in 2007 to 15 percent nationwide by 2022,

¹⁶The first universal mandate was adopted in Arizona in 2007. Appendix Section A.2 provides the full list of states with E-Verify mandates, the type of mandate, and the share of establishments enrolled in each state in 2022.

but reached 37 percent in states with universal E-Verify mandates. The top five states experienced much larger increases, with average enrollment of nearly 60 percent. As we can see, enrollment varies considerably across states and remains incomplete, even in states where universal mandates are in place.¹⁷ These observations motivate our counterfactual simulations: by varying enforcement intensity, we assess how different levels of E-Verify uptake affect labor market outcomes. This is particularly relevant in light of the ongoing policy debate over expanding E-Verify, where the consequences of stronger or weaker enforcement remain a key concern. After presenting the counterfactual results in Section 5, we interpret them using these numbers as empirical benchmarks (Section 5.4).

Figure 1: E-Verify Enrollment Rates for All Establishments by States



Notes: Figure 1 displays E-Verify enrollment as a fraction of the total number of establishments over time, for the U.S. as a whole, for states with universal E-Verify mandates, and for the five states with the highest adoption (Alabama, Arizona, Georgia, Missouri, and South Carolina). The data on the number of establishments enrolled in the E-Verify system by each state and year are obtained from the DHS via a Freedom of Information Act (FOIA) request. The data on the total number of establishments are obtained from the U.S. Census Bureau ([U.S. Census Bureau 2007–2022](#)).

¹⁷For example, among the states with universal mandates, the enrollment rate reached about 40 percent in Alabama and Arizona by late 2015, compared with only 10 percent in North Carolina. By contrast, public-sector-only mandates generated minimal uptake: the average enrollment rate in Colorado remained below 10 percent in 2015, nine years after its initial enactment. These patterns are consistent with those documented by [Orrenius, Zavodny, and Greer \(2020\)](#) and [Ayromloo, Feigenberg, and Lubotsky \(2020\)](#), who use similar DHS data.

3.2 Data

For calibration, we use individual-level data from the American Community Survey (ACS) (Ruggles et al. 2023), focusing on low-skilled workers—defined as individuals with a high school education or less—across all 50 states and the District of Columbia. As a large, nationally representative dataset with detailed demographic and labor market variables, the ACS provides repeated cross-sections with sufficient sample sizes to reliably measure labor market outcomes for small subpopulations—such as low-skilled workers and undocumented immigrants—compared to alternative sources like the Current Population Survey (CPS).¹⁸ Crucially, the ACS includes information on citizenship status, which allows us to construct proxies for likely undocumented individuals, as discussed in the next paragraph.

Among low-skilled workers, we identify *likely* undocumented immigrants as Hispanic non-citizens, following common practice in the literature (Passel and D’Vera Cohn 2009; Amuedo-Dorantes and Bansak 2014; Pope 2016; Cascio, Lewis, and Zhang 2024).¹⁹ This classification yields three mutually exclusive groups for analysis: likely undocumented immigrants, Hispanic natives, and non-Hispanic natives. While the latter two groups are not directly subject to immigration enforcement, they are likely to be close labor market substitutes and may be indirectly affected through labor market dynamics. Within each group, we further restrict the sample to individuals ages 21 to 40, since undocumented workers constitute more than 10 percent of the low-skilled sample in this range but not among those younger than 21 or older than 40.²⁰

Our primary outcomes—used as moment conditions in the calibration and counterfactual analyses—are unemployment rates and hourly wages for each of the three worker groups. Unemployment rates are defined as the share of unemployed individuals in the labor force within each group. Since the ACS does not report hourly wage directly, we follow the approach of East et al. (2023) and calculate hourly wages by dividing annual earnings by annual hours worked, where annual hours equal usual weekly hours multiplied by weeks worked per year. To reduce the influence of outliers, we winsorize wages at the 2.5th and 97.5th percentiles.²¹ Hourly wages are then averaged by group.

¹⁸Despite the small sample size of the CPS, the results remain qualitatively similar when we calibrate the model parameters using CPS moments.

¹⁹The residual imputation strategy of Borjas (2017) suggests that 72 percent of Hispanic immigrants are undocumented.

²⁰This pattern is not specific to our sample but reflects a broader demographic feature that undocumented workers are disproportionately concentrated among young adults (Millet and Paviol 2022). Our results are qualitatively similar when we lower the threshold for defining a salient share of undocumented workers from 10 percent to 5 percent, which expands the age range to 19–47. Once older or underage workers are included, however, the share of likely undocumented workers becomes negligible.

²¹Annual earnings refer to pre-tax wage and salary income of both full-time and part-time workers, adjusted for inflation to 2016 dollars. Weeks worked are reported in intervals, so we use the midpoint of each

Our final moment is the monthly unemployment-to-employment (UE) transition rate for low-skilled workers. The UE rate is defined as the share of workers who become employed in the following month among those unemployed in the previous month. Since the rotating panel structure is essential for constructing this measure, we use the Current Population Survey (CPS) (Flood et al. 2024). We construct a single measure for all low-skilled workers rather than for each subgroup in order to mitigate concerns about small sample size.

3.3 Model Extension

In this subsection, we adapt the baseline model from Section 2 to the context of E-Verify. In this setting, the observable characteristic is a worker’s ethnicity ($e \in \{\text{non-Hispanic } (nh), \text{Hispanic } (h)\}$), while the unobservable characteristic is their immigration status ($s \in \{\text{native } (n), \text{immigrant } (i)\}$). We assume that all non-Hispanic workers are natives and focus on Hispanic immigrants as undocumented workers.²² This yields three groups of workers: non-Hispanic natives (nh, n), Hispanic natives (h, n), and undocumented immigrants (h, i). We assume that all workers are identical except for their flow value of unemployment ($b_{e,s}$).

To model E-Verify, we adopt a parsimonious approach by introducing a key policy parameter in our model: the probability that a worker is flagged by the E-Verify system, denoted $\lambda_{e,s}$, which varies by immigration status. Workers are screened through E-Verify after matching with a firm but before wage bargaining, which closely mirrors actual procedures. If an undocumented worker is flagged, the match is voided—leaving the worker unemployed and the vacancy unfilled.

As we treat $\lambda_{e,s}$ as an exogenous policy variable capturing enforcement intensity, the interpretation is that states enforce E-Verify mandates through penalties and compliance checks, inducing $\lambda_{e,s}$ percent of firms to use the system. In practice, the probability of being flagged is the product of two components: the firm’s use of E-Verify and the system’s accuracy in identifying undocumented status. Given the system’s high accuracy, we assume that native workers are never incorrectly flagged ($\lambda_{nh,n} = \lambda_{h,n} = 0$), and interpret the probability that an undocumented worker is flagged ($\lambda_{h,i}$) as the system’s usage rate.²³

interval as the estimate. We exclude individuals who report zero hours or weeks worked, those are currently attending school, and those who are self-employed or unpaid family workers. Winsorization replaces values below the 2.5th percentile and above the 97.5th percentile with the respective threshold values.

²²This approach is based on the established link between Hispanic ethnicity and undocumented status among immigrants, as well as on standard practices in the literature for identifying likely undocumented immigrants (Passel and D’Vera Cohn 2009; Amuedo-Dorantes and Bansak 2014; Pope 2016; Borjas 2017; Cascio, Lewis, and Zhang 2024), as discussed in previous sections.

²³In Fiscal Year (FY) 2024, among employees who receive initial system mismatches, approximately 10 percent are later confirmed as work authorized after taking action to resolve the mismatch (i.e., false positives). Since legal workers have strong incentives to contest mismatches, we interpret the true error

Introducing the probability of matches flagged by the E-Verify system ($\lambda_{e,s}$) alters the value functions of unemployed workers and of vacancies. This change arises because a $\lambda_{h,i}$ fraction of meetings no longer result in successful matches. Only when a worker is not flagged do the firm and the worker engage in Nash bargaining over wages and form a match. The value function of an unemployed worker now becomes:

$$V_{e,s}^u = b_{e,s} + \beta \left\{ p(\theta) \frac{\psi_e}{\alpha_e} \left[(1 - \lambda_{e,s}) V_{e,s}^e(w(\bar{z}), \bar{z}) + \lambda_{e,s} V_{e,s}^u \right] + (1 - p(\theta)) \frac{\psi_e}{\alpha_e} V_{e,s}^u \right\} \quad (15)$$

The value function of a vacancy changes in a similar way:

$$0 = -k + \max_{\psi_e} \left\{ q(\theta) \sum_e \psi_e \left(\sum_s \gamma_{e,s} (1 - \lambda_{e,s}) V_{e,s}^f(w(\bar{z}), \bar{z}) \right) - \Omega \sum_e (\psi_e - \alpha_e)^2 \right\} \quad (16)$$

The value functions of an employed worker and a filled job remain unchanged.

From the firms' new maximization problem, we derive the optimal choice of tilting toward non-Hispanic workers, ψ_{nh} :

$$\psi_{nh}^* = \alpha_{nh} + \frac{q(\theta) \left[V_{nh,n}^f(w(\bar{z}), \bar{z}) - \gamma_{h,n} V_{h,n}^f(w(\bar{z}), \bar{z}) - \gamma_{h,i} (1 - \lambda_{h,i}) V_{h,i}^f(w(\bar{z}), \bar{z}) \right]}{4\Omega} \quad (17)$$

E-Verify usage creates a wedge that reduces the value of meeting with undocumented immigrant workers—all of whom are Hispanics in our setup—as some of these meetings no longer result in a match. This wedge directly increases firms' incentive to match with non-Hispanic workers to avoid the risk of failing to form a match after meeting a worker, thereby incurring the vacancy posting cost without any return. This leads to an increase in the discrimination ratio, referring to the ratio of the match probability for non-Hispanic workers to that for Hispanic workers.

However, general equilibrium effects can offset this increase in the discrimination ratio. First, the share of unemployed Hispanic workers who are undocumented, $\gamma_{h,i}$ increases with higher values of $\lambda_{h,i}$. This, in turn, leads firms to reduce their choice of ψ_{nh} . Second, the decline in the job finding rate for undocumented workers, resulting from the increase in $\lambda_{h,i}$, lowers their reservation productivity. This increases the match surplus associated with undocumented workers and thus reduces the optimal choice of ψ_{nh} .

rate as not much higher than this false positive rate. The data are available at: <https://www.e-verify.gov/about-e-verify/e-verify-data/e-verify-performance>.

4 Calibration

We calibrate the model to the labor market for low-skilled workers in the U.S. Table 1 reports the externally set parameters (Panel A), the targeted and model-implied moments (Panel B), and the estimated parameters (Panel C). We use outcomes from the three years prior to the first enactment of universal mandates in Arizona in 2007, corresponding to 2004-2006 for the entire U.S.

Panel A of Table 1 presents the externally calibrated parameters, including their values and sources. We set the measure of each worker type ($m_{e,s}$) to match the proportions observed in the ACS. Prior to the enforcement of E-Verify mandates, the low-skilled labor force consists of approximately 13.7 percent likely undocumented immigrant workers, 11.3 percent Hispanic native workers, and 75.1 percent non-Hispanic native workers. Match productivity shocks are assumed to be uniformly distributed between 0 and the productivity of a new match, denoted by $\bar{z}$. We adopt a Cobb-Douglas matching function of the form $AU^\varepsilon V^{1-\varepsilon}$, setting the elasticity parameter (ε) to 0.5 and the matching efficiency constant (A) to 0.113, following Chassamboulli and Peri (2015). The rate of exogenous separations (δ) is also set to 0.034, consistent with their calibration. Since we focus on the period before E-Verify enforcement, we set the E-Verify usage rate ($\lambda_{h,i}$) to zero.²⁴

We then jointly calibrate the remaining model parameters to match the unemployment rate ($u_{e,s}$) and average wage ($w_{e,s}$) for each worker type (i.e., non-Hispanic natives, Hispanic natives, and undocumented immigrants), as well as the monthly UE rate of low-skilled workers. The results are presented in Panel B of Table 1. Unemployment rates vary across groups, with undocumented workers facing a rate of 7.1 percent, compared to 10.7 percent for Hispanic natives and 11.3 percent for non-Hispanic natives. Average wages also differ by group: undocumented immigrants earn a mean hourly wage of \$13.4, while Hispanic native and non-Hispanic native workers earn \$14.7 and \$14.8, respectively. Compared to the targeted moments, the model does a good job of matching the moments from the ACS.

From the calibration shown in Panel C Table 1, we find that undocumented immigrant workers and Hispanic native workers have a lower value of home productivity relative to non-Hispanic native workers (i.e., $b_{h,i} < b_{h,n} < b_{nh,n}$). This leads firms to target their vacancies towards Hispanic workers. In the calibrated model, the discrimination ratio (i.e., the ratio of the match rate for non-Hispanic workers to that for Hispanic workers) is 0.89, indicating that firms choose to match with non-Hispanic workers about 10 percent less often than they would under fully random search.

²⁴Although employers had the option to use the E-Verify system prior to the enactment of state-level mandates, Orrenius, Zavodny, and Greer (2020) document that its usage rate was virtually zero before 2007.

Table 1: Baseline Parameterization

Parameter	Definition	Value	Source or Target
A. Externally Calibrated			
β	Discount rate	0.98	Standard
ε	Elasticity of the matching function	0.5	Chassamboulli and Peri (2015)
η	Worker's bargaining power	0.5	Chassamboulli and Peri (2015)
A	Coefficient on the matching function	0.113	Chassamboulli and Peri (2015)
$m_{nh,n}$	Measure of non-Hispanic natives	0.751	ACS Data
$m_{h,n}$	Measure of Hispanic natives	0.113	ACS Data
$m_{h,i}$	Measure of undocumented immigrants	0.137	ACS Data
δ	Exogenous separation rate	0.034	Chassamboulli and Peri (2015)
$\lambda_{h,i}$	E-Verify usage rate	0	Orrenius, Zavodny, and Greer (2020)
B. Targeted			
$u_{nh,n}$	Unemployment rate of non-Hispanic natives	11.3%	11.3%
$u_{h,n}$	Unemployment rate of Hispanic natives	10.7%	10.2%
$u_{h,i}$	Unemployment rate of undocumented immigrants	7.1%	9.8%
$w_{nh,n}$	Average wage of non-Hispanic natives	\$14.8	\$14.8
$w_{h,n}$	Average wage of Hispanic natives	\$14.7	\$14.7
$w_{h,i}$	Average wage of undocumented immigrants	\$13.4	\$13.4
UE	Monthly UE rate for all workers	0.39	0.39
C. Estimated			
k	Vacancy posting cost	0.32	
Ω	Coefficient on the cost of distorting match probabilities	3.88	
ϕ	Arrival rate of productivity shocks	0.016	
$\bar{z}$	Productivity of a new match	15.32	
$b_{nh,n}$	Home production value for non-Hispanic natives	11.48	
$b_{h,n}$	Home production value for Hispanic natives	10.33	
$b_{h,i}$	Home production value for undocumented immigrants	1.65	

Notes: Table 1 presents the parameters and targeted and matched moments prior to the enactment of E-Verify mandates. Panel A reports the parameters that are externally set from the literature and the ACS data. Panel B reports the targeted moments from the ACS and CPS data and the corresponding model-implied moments for unemployment rates and average wages across demographic groups, and the unemployment-to-employment (UE) rate. The UE rate is defined as the share of workers who become employed in the following month among those unemployed in the previous month. Panel C reports the parameters calibrated to match the moments from the ACS and CPS.

We also calculate the amount firms pay to target Hispanic workers as a fraction of the vacancy posting cost ($= 2\Omega(\psi_{nh} - \alpha_{nh})^2/k$). In the baseline calibration, we find that firms pay about 1 percent of the vacancy posting cost per vacancy to increase the rate at which

they match with Hispanic workers, suggesting that tilting costs are relatively small.

5 Counterfactual Analysis

In this section, we use our calibrated model to examine the effects of introducing E-Verify mandates. The most direct effect is an increase in the usage rate of E-Verify, captured by an increase in $\lambda_{h,i}$. Another potentially important margin of adjustment is that undocumented workers might migrate out of states with E-Verify mandates in search of better employment opportunities or a less hostile environment toward undocumented immigrants. In our model, this is reflected by a decrease in the measure of undocumented workers $m_{h,i}$.²⁵

Thus, we explore the effects of the mandates by solving our model under different values of the E-Verify usage rate ($\lambda_{h,i} \in [0, 1]$) and the out-migration rate of undocumented workers ($o \in [0, 0.6]$, where $m_{h,i}^o = (1 - o)m_{h,i}$). Specifically, we begin by analyzing how E-Verify affects firms' targeting behavior toward Hispanic versus non-Hispanic workers (Section 5.1), as well as the resulting impacts on employment and wages (Section 5.2). We then assess the welfare implications of E-Verify mandates (Section 5.3). Lastly, we provide empirical benchmarks on the observed increases in E-Verify usage and out-migration of undocumented workers following the introduction of E-Verify mandates, and discuss the quantitative implications based on our model (Section 5.4).

5.1 Firm Tilting

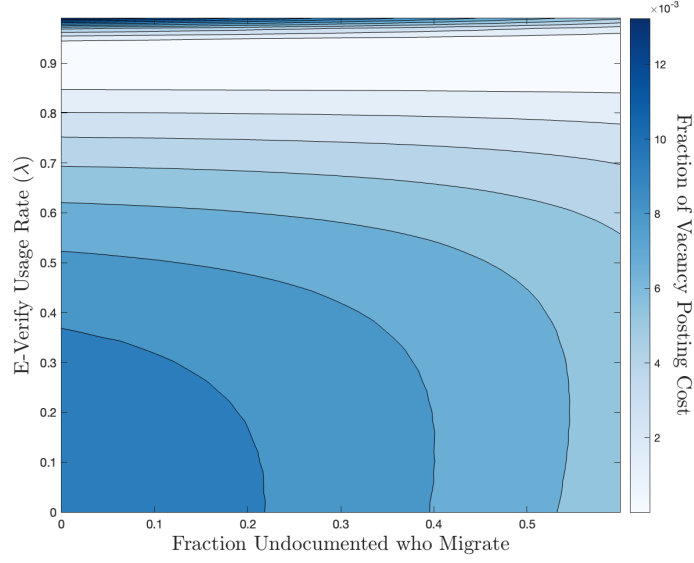
We first examine the effects of E-Verify enforcement on firm's tilting toward different worker types, i.e., the extent to which firms are deviating from random search. Figure 2 plots the tilting costs paid by firms as a fraction of the vacancy posting cost ($= 2\Omega(\psi_{nh} - \alpha_{nh})^2/k$) (Figure 2-A) and the discrimination ratio ($= \frac{\psi_{nh}\alpha_h}{\alpha_{nh}\psi_h}$) (Figure 2-B), across varying levels of E-Verify usage ($\lambda_{h,i}$) and out-migration rate of undocumented workers (o).

In Figure 2-A, the tilting costs paid by firms (as a share of the vacancy posting cost) initially decline as E-Verify usage rises (i.e., increase in $\lambda_{h,i}$), since firms reduce their targeting of Hispanic workers. At high levels of $\lambda_{h,i}$, however, firms start actively targeting non-Hispanic workers, causing tilting costs to rise again, as they are willing to pay to avoid matches with Hispanic workers. This pattern arises because higher E-Verify usage reduces

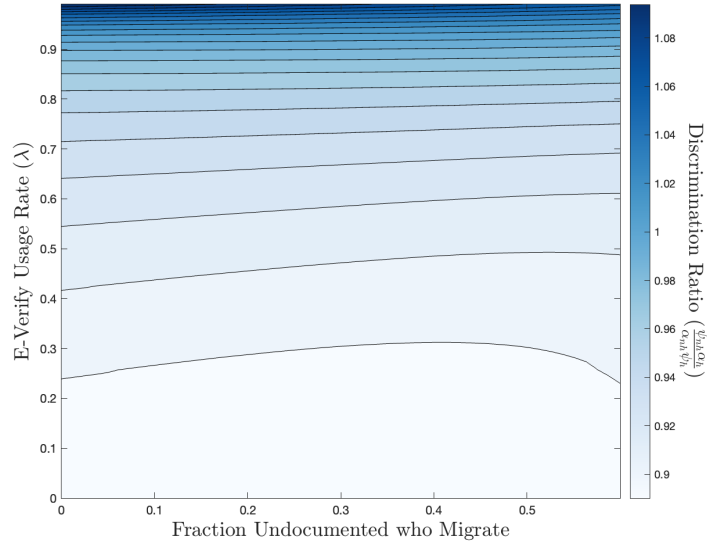
²⁵Previous empirical studies as well as data we collect from the DHS provide evidence that E-Verify mandates led to an increase in firms' enrollment in the system (Ayromloo, Feigenberg, and Lubotsky 2020; Orrenius, Zavodny, and Greer 2020) and the out-migration of undocumented immigrants (Bohn, Lofstrom, and Raphael 2014; Orrenius and Zavodny 2016; Churchill 2021).

Figure 2: Counterfactual Analysis: Tilting Cost and Discrimination Ratio

A. Tilting Cost



B. Discrimination Ratio



Notes: Figure 2 displays the change in the tilting costs paid by firms (as a share of the vacancy posting cost) (Figure 2-A) and the discrimination ratio (Figure 2-B) across different values of the E-Verify usage rate ($\lambda_{h,i}$) and the share of undocumented workers who migrate out.

the likelihood of meetings with undocumented workers resulting in matches, lowering the expected value of targeting Hispanic workers.

Likewise, increased out-migration of undocumented workers (i.e., increase in o and thus decrease in $m_{h,i}$) reduces tilting costs, as firms shift away from targeting Hispanic workers. This occurs because out-migration reduces the likelihood that a Hispanic worker is undocumented and thus has a lower home production value (i.e., outside option). Moreover, if increases in E-Verify usage lead to out-migration of undocumented workers, the effects on the tilting costs paid by firms are stronger than if no such out-migration occurred. This is simply due to the fact that the effects of both increased E-Verify usage and increased out-migration of workers work in the same direction.

Turning to Figure 2-B, the discrimination ratio—which captures the relative match rates between non-Hispanic and Hispanic workers—rises with the E-Verify usage rate, increasing from 0.89 in the baseline calibration without E-Verify usage ($\lambda_{h,i} = 0$) to 1.1 under strong enforcement ($\lambda_{h,i} \approx 1$). This indicates that the match rate for non-Hispanic workers, initially about 10 percent lower than that for Hispanic workers, becomes approximately 10 percent higher under full enforcement. As undocumented workers migrate out, the discrimination ratio decreases slightly, meaning that the match rate for Hispanic workers increases relative to that for non-Hispanic workers. This occurs even though firms target Hispanic workers *less*, because the proportion of Hispanics in unemployment decreases more than the decline in firms’ tilting towards them.

5.2 Unemployment and Wages

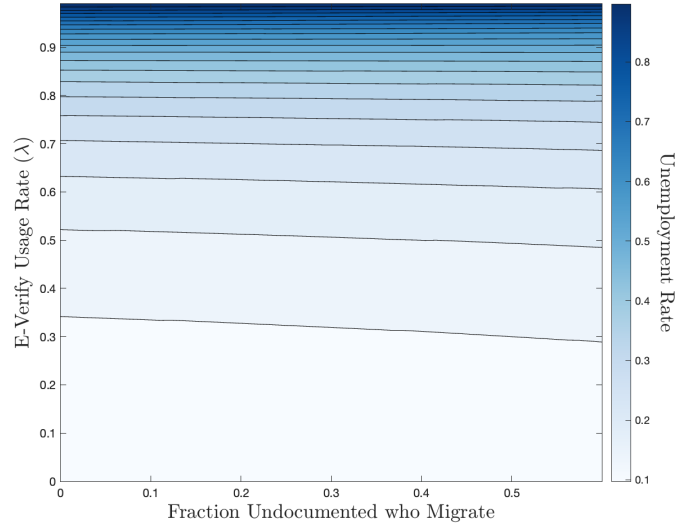
Next, we turn to the effects of E-Verify mandates on labor market outcomes for the three worker types: unemployment rates (Figure 3) and average wages (Figure 4) for undocumented immigrant workers, non-Hispanic native workers, and Hispanic native workers. Unlike undocumented workers, native workers are not directly subject to enforcement. Still, examining the impacts on natives, separately by their Hispanic ethnicity, provides a complete understanding of how E-Verify enforcement affects firms’ behavior and labor market dynamics in the low-skilled sector.

For undocumented workers (Figure 3-A), who are directly impacted by E-Verify enforcement, an increase in the usage rate of E-Verify leads to a sharp increase in unemployment, with the unemployment rate exceeding 80 percent at high levels of $\lambda_{h,i}$. This is due to the direct effect of E-Verify, which prevents some meetings from resulting in matches, compounded by reduced tilting of vacancy postings toward Hispanic workers.

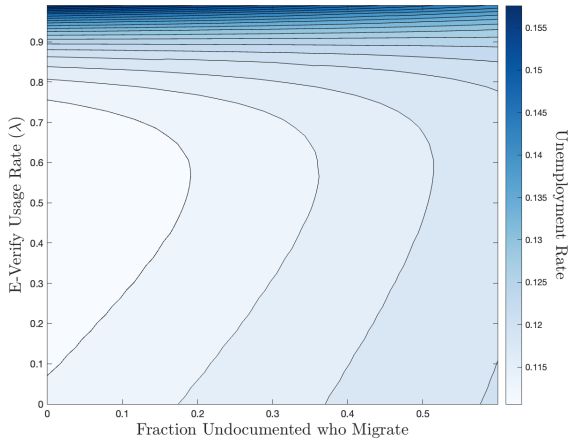
As another dimension, an increase in the out-migration of undocumented workers also

Figure 3: Counterfactual Analysis: Unemployment Rates

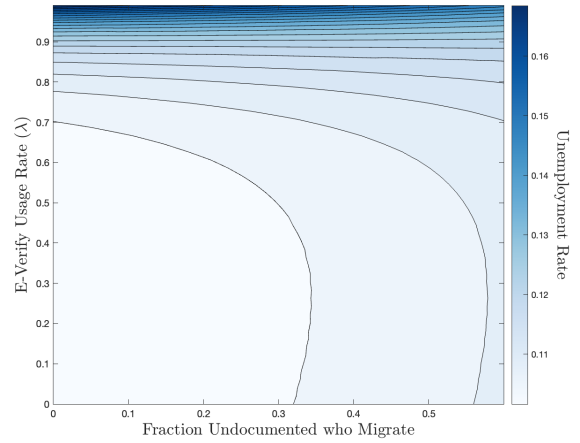
A. Undocumented Immigrants



B. Non-Hispanic Natives



C. Hispanic Natives



Notes: Figure 3 displays the change in unemployment rates across different values of the E-Verify usage rate ($\lambda_{h,i}$) and the share of undocumented workers who migrate out, shown separately for each worker type.

results in higher unemployment. Specifically, the unemployment rate increases by 0.7 percentage points relative to the baseline when 60 percent of undocumented workers migrate out. This is driven by a reduction in the average value of posting a vacancy, which lowers labor market tightness and the match rate between firms and workers. As a result, the increase is comparable to that of native workers, since there is no mechanism uniquely targeting undocumented workers in this case.

Although native workers are not the direct target of E-Verify, our results demonstrate that the enforcement has spillover effects on them. The unemployment rate for non-Hispanic native workers (Figure 3-B) exhibits a slight U-shape with respect to the E-Verify usage rate ($\lambda_{h,i}$). This pattern is driven by two offsetting effects of higher $\lambda_{h,i}$ on the labor market tightness. On the one hand, an increase in $\lambda_{h,i}$ reduces the probability of matching with high-surplus undocumented workers, which lowers the value of posting a vacancy. On the other hand, an increase in $\lambda_{h,i}$ reduces the extent of vacancy tilting by firms, thereby lowering the average cost of posting a vacancy and encouraging more vacancy creation.

For smaller values of $\lambda_{h,i}$, the effects of reduced vacancy posting costs dominate, which stems from less vacancy tilting along with the decreased targeting of Hispanic workers. This leads to an increase in the match rate for non-Hispanic workers and a decrease in their unemployment rate by up to 0.3 percentage points relative to the baseline for $\lambda_{h,i}$ of around 0.55. However, for larger values of $\lambda_{h,i}$ the reduction in the average value of workers to the firm dominates, combined with increasing tilting costs. This results in a sharp decline in the match rate for non-Hispanic workers and an increase in their unemployment rate up to 4.7 percentage points relative to the baseline. This occurs despite the fact that firms are now actively targeting non-Hispanic workers.

For Hispanic native workers (Figure 3-C), the unemployment rate initially decreases as E-Verify usage rises—though less than for non-Hispanic natives—but then increases more sharply than for non-Hispanic natives under higher usage. This difference arises because Hispanic natives share an observable characteristic (i.e., Hispanic ethnicity) with undocumented workers, which firms must rely on prior to meeting the worker since they cannot directly observe a worker’s legal status. Although labor market tightness initially rises, this is offset by a reduction in firms’ tilting toward Hispanic workers. As $\lambda_{h,i}$ increases further, Hispanic workers face the combined effects of declining labor market tightness and reduced targeting toward Hispanic workers by firms, leading to even larger increases in unemployment. For large values of $\lambda_{h,i}$, the unemployment rate for Hispanic natives rises by up to 7 percentage points relative to the baseline.

Turning to the effects of out-migration of undocumented workers, we find that E-Verify increases unemployment among all native workers (Figures 3-B and 3-C). The same two

forces are at play with increases in o (and thus decreases in $m_{h,i}$) as with increases in $\lambda_{h,i}$: a decrease in the share of undocumented workers lowers both the average value and the average cost of posting a vacancy. However, unlike increases in $\lambda_{h,i}$, an increase in o lowers the average value of a worker to an extent that outweighs the reduction in vacancy posting costs from less tilting. Even so, both effects are more modest than under increases in $\lambda_{h,i}$, with increases in unemployment rates of about 1 percentage point when 60 percent of undocumented workers migrate out ($o = 0.6$). This is because in this case, vacancy posting costs are never lost due to enforcement.

Finally, the negative effects of the out-migration of undocumented workers on the unemployment rates of native workers are reversed for very large values of E-Verify usage ($\lambda_{h,i} > 0.9$). In this case, the out-migration of undocumented workers lowers the unemployment rate of native workers, as it reduces the frequency with which the vacancy posting costs are incurred without resulting in a match.

Next, Figure 4 plots the average wages for each type of worker across different levels of E-Verify usage and out-migration of undocumented workers. The patterns for wages mirror those observed for unemployment, though with the opposite signs. As the E-Verify usage rate increases, wages decline for both undocumented workers (Figure 4-A) and Hispanic native workers (Figure 4-C). The wages of non-Hispanic workers (Figure 4-B), on the other hand, rise slightly at first but then decline as $\lambda_{h,i}$ increases further. Wages for all worker types also decrease with out-migration of undocumented workers.

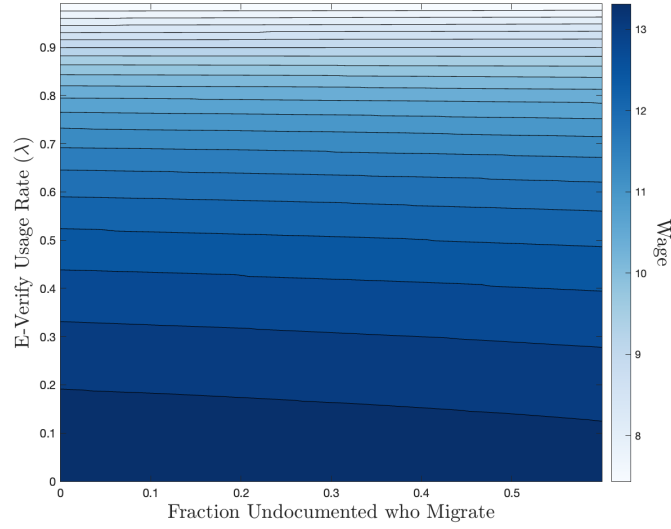
While the wage effects for native workers (Figures 4-B and 4-C) are relatively modest—even at high levels of E-Verify usage ($\lambda_{h,i} > 0.9$), wages decline by only about 1 to 2 percent—the effects for undocumented workers (Figure 4-A) are much larger, with average wages falling by nearly 50 percent. Furthermore, as with unemployment rates, the negative effects of the out-migration of undocumented workers on native workers’ wages are reversed when E-Verify usage is very high.

Our results from the counterfactual analyses are consistent with several strands of the literature on the labor market effects of immigration enforcement. First, they align with dynamic search-and-matching models without information frictions, which conclude that E-Verify enforcement can worsen labor market outcomes not only for undocumented workers but also for native workers due to the dominant effect of reduced vacancy postings by firms (Chassamboulli and Palivos 2014; Chassamboulli and Peri 2015; Albert 2021). Although focused on a different type of immigration enforcement, the simulation by Albert (2021) finds negative effects on low-skilled natives.²⁶

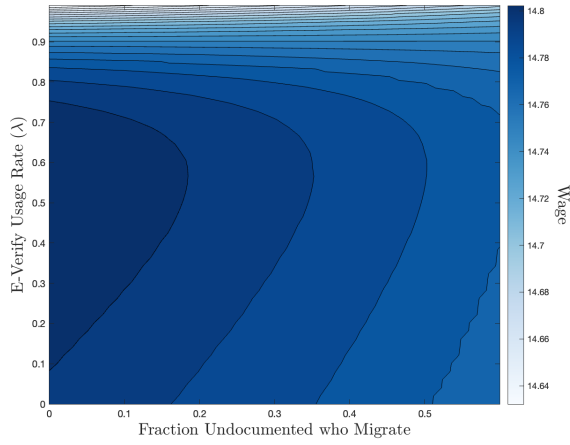
²⁶They simulate the impact of a counterfactual policy involving stricter immigration enforcement through an increase in the deportation rate of undocumented immigrants, whereas we focus on changing the proba-

Figure 4: Counterfactual Analysis: Wages

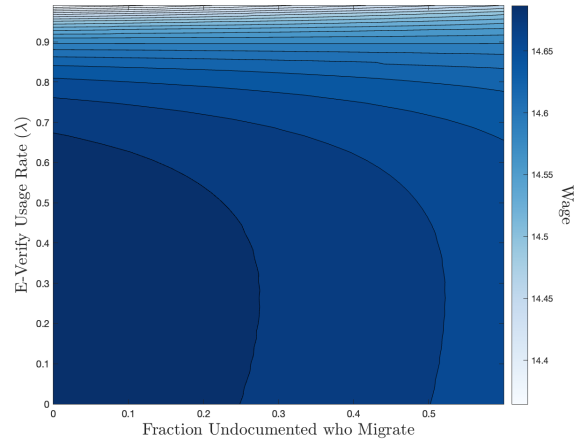
A. Undocumented Immigrants



B. Non-Hispanic Natives



C. Hispanic Natives



Notes: Figure 4 displays the change in average wages across different values of the E-Verify usage rate ($\lambda_{h,i}$) and the share of undocumented workers who migrate out, shown separately for each worker type.

The results also align with the most recent reduced-form evidence on the adverse effects of E-Verify mandates on the labor market outcomes of natives, which is most comparable to our setting (Ayromloo, Feigenberg, and Lubotsky 2020).²⁷ Moreover, the adverse effects of E-Verify are amplified by firm’s tilting behavior based solely on ethnicity, leading to particularly negative consequences for Hispanic natives. Although not directly comparable, Ayromloo, Feigenberg, and Lubotsky (2020) also document that the negative effects of E-Verify on low-skilled workers are more pronounced among Hispanic workers.²⁸ They further show that the adverse effects on native employment are driven almost entirely by younger workers, who overlap more with the undocumented workforce and are the focus of our analysis.

In sum, our model not only replicates existing empirical evidence on the effects of immigration enforcement but also introduces a new, quantitatively relevant mechanism in a parsimonious way.

5.3 Welfare

In this subsection, we explore the welfare implications of E-Verify enforcement, focusing on changes in its usage and the out-migration of undocumented workers. We consider two different welfare criteria: total welfare per worker in the economy (W_a) and welfare per worker for native workers specifically (W_n). Since we solve the model in steady state, we focus on welfare in a single period. Total welfare in the economy is therefore given by:

bility of failed matches between workers and firms, which more directly affects the matching process. Despite this difference, their analysis yields similar predictions about the effects of employment enforcement. First, the negative impact on natives’ employment outcomes is larger when deportation policies primarily target undocumented immigrants at the workplace, rather than being applied regardless of employment status. Second, the adverse effects on natives are greater when penalties are imposed on firms that hire unauthorized workers, as this further discourages vacancy creation.

²⁷They use the same ACS data as we do and focus exclusively on the effects of universal E-Verify mandates. Following the mandate enactment, they find a 17.5 to 19 percent decline in the employment rate among likely undocumented immigrants and a 2.7 percent decline among low-skilled natives.

²⁸A direct one-to-one comparison with their estimates is challenging for several reasons. First, our primary variable of interest is the fraction of unemployed workers within the labor force, rather than within the entire working-age population. Second, our classification of worker groups does not exactly correspond to theirs. Using ACS data, they examine heterogeneity by gender and age within the native population but not by Hispanic ethnicity. They examine heterogeneity by Hispanic ethnicity using another dataset named the Quarterly Workforce Indicators (QWI) and find larger adverse effects on employment among all Hispanics (an 8.7% decline) compared to non-Hispanics (a 2.9% decline). These estimates reflect outcomes for both documented and undocumented workers, as the QWI provides aggregate labor market measures with limited demographic information on workers.

$$\begin{aligned}
W_a = & -(k + 2\Omega(\psi_{nh} - \alpha_{nh})^2)\theta u + b_{nh,n}u_{nh,n} + b_{h,n}u_{h,n} + b_{h,i}u_{h,i} \\
& + (m_{nh,n} - u_{nh,n}) \int_{R_{nh,n}}^{\bar{z}} z dG_{nh,n}(z) \\
& + (m_{h,n} - u_{h,n}) \int_{R_{h,n}}^{\bar{z}} z dG_{h,n}(z) \\
& + (m_{h,i} - u_{h,i}) \int_{R_{h,i}}^{\bar{z}} z dG_{h,i}(z)
\end{aligned} \tag{18}$$

which we then divide by the total measure of workers in the economy ($= m_{nh,n} + m_{h,n} + m_{h,i}$) to get welfare per worker, where u is the total measure of unemployed workers ($u = u_{nh,n} + u_{h,n} + u_{h,i}$). The first term in the equation represents the vacancy posting cost paid by firms in a given period; the second term captures the vacancy tilting cost paid by firms; and the remaining terms represent the sum of home productivity and labor productivity across all worker types.

Similarly, we define welfare for native workers as follows:

$$\begin{aligned}
W_n = & -(k + 2\Omega(\psi_{nh} - \alpha_{nh})^2)\theta u + b_{nh,n}u_{nh,n} + b_{h,n}u_{h,n} \\
& + (m_{nh,n} - u_{nh,n}) \int_{R_{nh,n}}^{\bar{z}} z dG_{nh,n}(z) \\
& + (m_{h,n} - u_{h,n}) \int_{R_{h,n}}^{\bar{z}} z dG_{h,n}(z)
\end{aligned} \tag{19}$$

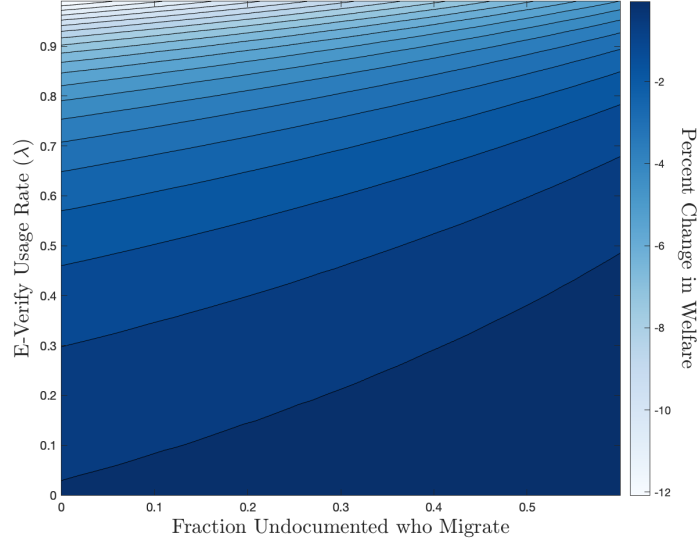
which we divide by the total number of native workers in the economy ($= m_{nh,n} + m_{h,n}$) to get welfare per native worker. This measure is identical to total welfare in Equation 18 but excludes the home productivity and labor productivity of undocumented workers.

Figure 5 plots the change in total welfare per worker and welfare per worker for native workers across different levels of E-Verify usage and out-migration of undocumented workers. In Figure 5-A, total welfare per worker in the economy increases only marginally with the out-migration of undocumented workers. Although unemployment rises across all groups—reducing aggregate productivity—this negative effect is offset by two forces. First, lower labor market tightness reduces the total cost to the firm of filling vacancies (i.e., the sum of the vacancy posting cost and the tilting cost). Second, the composition of the workforce shifts toward native workers, who have higher average welfare due to greater home productivity.

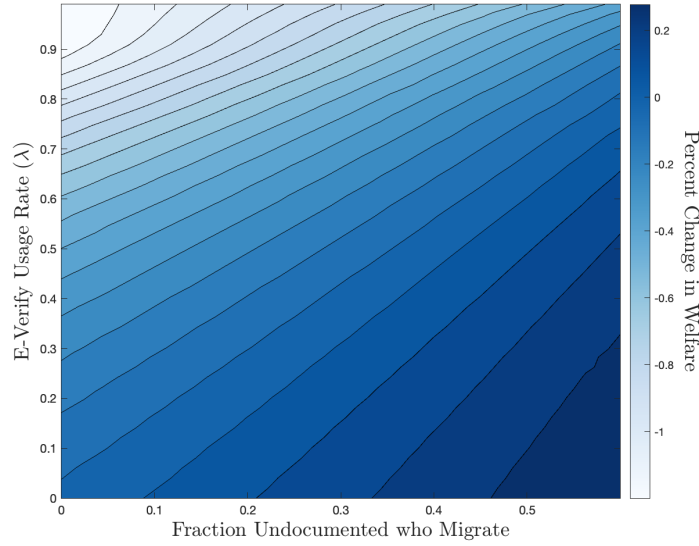
In contrast, increases in E-Verify usage reduce welfare per worker for two reasons. First, they cause some meetings between firms and workers to fail to form matches, which in

Figure 5: Counterfactual Analysis: Welfare

A. Total Welfare



B. Welfare of Native Workers



Notes: Figure 5 displays the change in total welfare per worker in the economy (Figure 5-A) and welfare per worker for native workers (Figure 5-B) across different values of the E-Verify usage rate ($\lambda_{h,i}$) and the share of undocumented workers who migrate out.

turn increases the likelihood that a firm incurs the vacancy posting cost without filling the position. Second, unlike decreases in $m_{h,i}$, increases in $\lambda_{h,i}$ do not affect the composition of workers in the labor force, so there is no welfare gain from a shift toward native workers with higher home productivity.

In Figure 5-B, we observe similar patterns when measuring the change in welfare per native worker across different levels of E-Verify usage and out-migration of undocumented workers: it rises slightly with undocumented out-migration and falls with higher E-Verify usage. However, these effects are smaller in magnitude than the overall impacts on total welfare. This is because native workers are not directly affected by enforcement, but only indirectly through spillover effects from the changes in firm tilting and vacancy creation.

5.4 E-Verify Usage and Out-Migration: Empirical Benchmarks

A key takeaway from our previous subsections is that the effects of E-Verify mandates depend critically on both the induced usage rate and the extent of undocumented worker out-migration. In this subsection, we explore the impacts on the low-skilled labor market predicted by our model in response to the increases in E-Verify usage discussed in Section 3.1 (Figure 1).²⁹

We focus on the effects of the observed increase in E-Verify usage in two groups: (i) states with universal mandates, where usage rose from 0 to 37 percent, and (ii) the five states with the largest increase—Alabama, Arizona, Georgia, Missouri, and South Carolina—where usage rose from 0 to about 60 percent. In our model, an increase in E-Verify usage to 37 percent ($\lambda_{h,i} = 0.37$), assuming no out-migration of undocumented workers, leads to modest increases in the unemployment rate (around 3.3 percentage points) and decreases in wages (around 3.8%) for undocumented workers. This change also results in small decreases in unemployment rates (around 0.1 percentage points) and small increases in wages (around one cent) for native workers. Furthermore, overall welfare per worker in the economy decreases by 0.7 percent and welfare per native worker decreases by 0.2 percent.³⁰

However, increases of E-Verify usage past 37 percent to 60 percent ($\lambda_{h,i} = 0.6$) exacerbate the negative effects and flip the positive effects. In our model, an increase in E-Verify usage to 60 percent leads to a decrease in the unemployment rate of non-Hispanic natives by 0.2

²⁹Although it provides a useful benchmark, the enrollment rate is not a strict one-to-one counterpart of the E-Verify usage rate in our model ($\lambda_{h,i}$) for several reasons. First, a firm’s enrollment in the E-Verify system does not necessarily imply actual usage, so enrollment rates may overstate usage. Conversely, if enrollment is concentrated among large firms—for instance, due to disproportionate compliance checks—the enrollment rate could understate the actual usage among new hires.

³⁰Welfare for native workers declines despite their higher wages and employment rate due to an increase in vacancy posting costs paid by firms.

percentage points, an increase in that of Hispanic natives by 0.1 percentage points, and an increase in that of undocumented workers by 11 percentage points. The average hourly wages of native non-Hispanics (Hispanics) increase (decrease) by about one cent, while those of undocumented workers fall by 1.5 dollars (an 11% decline). Overall welfare per worker in the economy decreases by 2.1 percent, and welfare per worker among native workers decreases by 0.54 percent. Stronger E-Verify enforcement beyond this level—resulting in usage rates above 70 percent—would lead to higher unemployment and lower wages not only for undocumented immigrants but also for all native workers, as well as larger adverse effects on welfare.

With regards to the out-migration elasticity, [Bohn, Lofstrom, and Raphael \(2014\)](#) find that the enactment of the E-Verify mandate in Arizona—accompanied by a range of other immigration enforcement measures—led to the out-migration of about 2 percent of undocumented workers in 2008 and 2009, the years immediately following the mandate’s implementation. During this period, E-Verify enrollment in Arizona increased from nearly 0 percent to about 25 percent, suggesting that a 1 percentage point increase in E-Verify usage rate resulted in the out-migration of roughly 0.08 percent of undocumented workers.

Based on the large increases in E-Verify enrollment in some states observed in our data (i.e., from 0% up to 60% in Alabama, Arizona, Georgia, Missouri, and South Carolina), we would expect about 5 percent of undocumented workers to migrate out of these states (an increase in ϕ from 0 to 0.05). Such small levels of out-migration have little impact on wages, unemployment rates, and worker welfare. This suggests that the responsiveness of out-migration to E-Verify usage, at least at lower levels, is quite limited and does not substantially affect labor market outcomes. However, if the responsiveness of out-migration to E-Verify usage were larger or non-linear, it could play an important role in shaping policy outcomes.

In sum, we find that large increases in E-Verify usage worsen labor market outcomes for low-skilled workers, including native workers, by reducing wages, employment rates, and overall welfare. Although native workers are not directly subject to E-Verify, they are negatively affected by spillover effects from reduced labor market tightness and changes in firms’ vacancy tilting behavior. These adverse effects are more severe for Hispanic native workers, who, in the absence of E-Verify, benefit from being observationally similar to undocumented workers. As enforcement intensity rises, this advantage diminishes due to reduced targeting, leading to worse outcomes for native Hispanics than for native non-Hispanics. Small increases in E-Verify usage slightly improve the labor market conditions of native workers but still reduce welfare for both undocumented and native workers. This occurs because vacancy posting costs are lost when matches are prevented from forming due to E-Verify.

6 Conclusion

Search and information frictions are defining features of labor markets. This paper introduces a novel mechanism of firm behavior under such frictions: firms can choose the probability that a matched worker belongs to a particular group based on their observable (but not unobservable) type. In doing so, firms statistically discriminate by tilting search toward groups perceived to yield a higher match surplus.

As an extension, we apply the model to U.S. immigration enforcement policy, specifically the E-Verify mandate, which aims to reduce unauthorized employment. Calibrated to match labor market outcomes for low-skilled workers, the model shows that strong E-Verify enforcement reduces job finding and job filling rates and distorts firms' search incentives. As a result, unemployment rises and wages fall not only for undocumented immigrants but also for native workers. Since firms target Hispanic workers in the absence of E-Verify in order to match more frequently with undocumented workers—who have a higher match surplus—once E-Verify usage increases, the adverse effects are larger for Hispanic than for non-Hispanic native workers. While out-migration of undocumented workers can mitigate the negative welfare effects of increased E-Verify usage, doing so would require unrealistically high levels of out-migration.

Our analysis provides a salient example of how policies targeting one subgroup of workers can generate spillover effects to untargeted groups of workers through employer behavior and labor market dynamics in the presence of informational frictions. Within the context of the low-skilled labor market, the message for policymakers is clear: expanding verification requirements without considering employer behavior can backfire, raising hiring costs and hurting the very workers the policy is meant to protect. As calls for nationwide expansion of E-Verify mandates grow in the U.S. and similar systems spread abroad, it is important to recognize these unintended consequences.

Appendix

A.1 Derivation of Optimal Tilting

This section presents the derivation of the firm's optimal tilting behavior, characterized by the choice of ψ_i as shown in Equation 5. Specifically, ψ_i is defined as the probability of meeting a worker of observable type $i = 1, \dots, I$ (conditional on having met a worker). Firms choose ψ_i to maximize the expected value of matching with a worker, which may differ from the fraction of unemployed workers of type i (denoted α_i) in our model, subject to the constraint $\sum_i \psi_i = \sum_i \alpha_i = 1$.

Using the value function of a vacancy in Equation 4 and the constraint $\sum_i \psi_i = 1$, we solve the following optimization problem for ψ_i :

$$0 = -k + \max_{\psi_i} \left\{ q(\theta) \sum_i \psi_i \left(\sum_j \gamma_{i,j} V_{i,j}^f(w(\bar{z}), \bar{z}) \right) - \Omega \sum_i (\psi_i - \alpha_i)^2 \right\} \quad \text{subject to} \quad \sum_i \psi_i = 1$$

Let λ be the Lagrange multiplier for the constraint $\sum_i \psi_i = 1$. Defining $A_i := \sum_j \gamma_{i,j} V_{i,j}^f(w(\bar{z}), \bar{z})$, the Lagrangian becomes:

$$\mathcal{L} = q(\theta) \sum_i \psi_i A_i - \Omega \sum_i (\psi_i - \alpha_i)^2 + \lambda \left(1 - \sum_i \psi_i \right)$$

Taking the derivative with respect to each ψ_i :

$$\frac{\partial \mathcal{L}}{\partial \psi_i} = q(\theta) A_i - 2\Omega(\psi_i - \alpha_i) - \lambda = 0$$

Solving for ψ_i :

$$\psi_i = \alpha_i + \frac{1}{2\Omega} (q(\theta) A_i - \lambda) \tag{20}$$

Imposing the constraint $\sum_i \psi_i = \sum_i \alpha_i = 1$:

$$\begin{aligned}
1 &= \sum_i \psi_i = \sum_i \left[\alpha_i + \frac{1}{2\Omega} (q(\theta)A_i - \lambda) \right] \\
&= \sum_i \alpha_i + \frac{1}{2\Omega} \left(q(\theta) \sum_i A_i - I\lambda \right) \\
&= 1 + \frac{1}{2\Omega} \left(q(\theta) \sum_i A_i - I\lambda \right)
\end{aligned}$$

Solving for λ :

$$\begin{aligned}
q(\theta) \sum_i A_i - I\lambda &= 0 \\
\lambda &= \frac{q(\theta)}{I} \sum_i A_i
\end{aligned}$$

Substituting λ into the expression for ψ_i in Equation 20 yields:

$$\begin{aligned}
\psi_i^* &= \alpha_i + \frac{1}{2\Omega} \left(q(\theta)A_i - \frac{q(\theta)}{I} \sum_{i'} A_{i'} \right) \\
&= \alpha_i + \frac{q(\theta)}{2\Omega} \left(A_i - \frac{1}{I} \sum_{i'} A_{i'} \right) \\
&= \alpha_i + \frac{q(\theta)}{2\Omega} \left(\sum_j \gamma_{i,j} V_{i,j}^f(w(\bar{z}), \bar{z}) - \frac{1}{I} \sum_{i'} \sum_j \gamma_{i',j} V_{i',j}^f(w(\bar{z}), \bar{z}) \right) \\
&= \alpha_i + \frac{q(\theta)}{2I\Omega} \sum_{i'} \sum_j \left(\gamma_{i,j} V_{i,j}^f(w(\bar{z}), \bar{z}) - \gamma_{i',j} V_{i',j}^f(w(\bar{z}), \bar{z}) \right)
\end{aligned}$$

which is equivalent to the expression presented in Equation 5.

A.2 States Implementing E-Verify Mandates

State	Enactment Date	Type of the Mandate	Enrollment Rate (in 2022)
Alabama	Jun, 2011	Universal	0.711
Arizona	Jul, 2007	Universal	0.486
Colorado	Jun, 2006	Public	0.125
Florida	Jan, 2011	Public	0.143
Georgia	Apr, 2006	Public	0.586
	May, 2011	Universal	
Idaho	May, 2009	Public	0.062
Indiana	May, 2011	Public	0.139
Louisiana	Jul, 2011	Universal	0.129
Michigan	Jun, 2012	Public	0.093
Minnesota	Jan, 2008	Public	0.091
Mississippi	Mar, 2008	Universal	0.246
Missouri	Jul, 2008	Public	0.531
Nebraska	Apr, 2009	Public	0.197
North Carolina	Aug, 2006	Public	0.163
	Jun, 2011	Universal	
Oklahoma	May, 2007	Public	0.112
Pennsylvania	Jul, 2012	Public	0.088
Rhode Island	Mar, 2008	Public	0.142
South Carolina	Jun, 2008	Public	0.592
	Jun, 2011	Universal	
Tennessee	Jun, 2011	Universal	0.178
Texas	Jun, 2015	Public	0.119
Utah	Mar, 2008	Public	0.180
	Mar, 2010	Universal	
Virginia	Apr, 2010	Public	0.141
West Virginia	Mar, 2012	Public	0.053

Notes: The last column reports the share of establishments enrolled in E-Verify in 2022. The law was revoked in January 2011 in Rhode Island.

Source: National Conference of State Legislatures (NCSL), U.S. Census Bureau ([U.S. Census Bureau 2007–2022](#)), and Freedom of Information Act (FOIA) requests

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